

The Long Asset

Constitutional architecture for a UK sovereign endowment

Drew Griffiths

*This paper states the financial findings on which its constitutional architecture depends. The companion paper, *The Arithmetic of Patience*, provides the full historical tests, methods, sensitivity analysis and falsification conditions underlying those findings. A reader need not read the companion to follow the argument here; it exists so that the financial proposition can be examined, and if necessary rejected, independently of the constitutional design.*

If the financial proposition fails there, the institution described here should not be created. A reader may accept the constitutional design and reject the modelling, or the reverse.

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Summary

The United Kingdom's pension expenditure has been rising for decades and is projected to keep rising, and there is no permanent public financial asset intended to meet it. This paper asks what institution could hold such an asset, and survive holding it long enough for the asset to matter.

The object is the state's net position, not the pension bill. A fund of this kind might contribute in two ways: by distributing an income against pension expenditure, or by converting borrowing into a growing public asset that improves the state's balance sheet. The companion paper, *The Arithmetic of Patience*, tests both against the historical record and finds that the income objective fails at any plausible contribution path while the balance sheet objective survives on the evidence tested. This paper therefore designs an institution for the second, and if the first is what a reader wants, the arithmetic does not support it.

The source of finance determines what the institution can be. Under debt financing only the balance sheet objective remains available, and only under a low distribution rule; genuine surplus financing makes both available and dissolves the trade-off. A receipt that would otherwise have reduced borrowing is debt financing whatever it is called. The funding question is therefore an architectural constraint rather than a menu.

The distribution rule is a constitutional choice, not an operational parameter. The paper adopts a rule of up to 5% of realised real return, never a share of principal, with nothing distributed before year 20. That rule is what keeps the institution a balance sheet institution rather than allowing it to become, by increments and without a decision, the income fund the arithmetic rejects.

Protection of principal is not protection of the contribution. The comparative record shows funds whose capital was legally secure and whose contributions were suspended, redirected or diluted, and the failures have different causes rather than one. Protection therefore cannot be reduced to a bar on withdrawal, and the architecture here separates investment management, fiscal contribution, temporary suspension and ultimate reversion, reserving automatic consequences for facts that can be determined without interpreting investment performance.

The central conclusion is institutional rather than predictive. A long horizon endowment is economically plausible on the evidence tested; whether it is achievable depends on protecting a decades long accumulation process from much shorter political and fiscal cycles.

The weakest link is named rather than solved. The fund must be defended during decades in which its cost is obvious and its benefit is not, and the paper proposes a mechanism for that without establishing that the mechanism works. It is the least evidenced element of the architecture and the only one addressing a failure mode the others cannot reach. The evidence that would settle it is behavioural rather than financial, and it will not resolve the way the modelling does.

What is not claimed. That a fund begun now could relieve pension expenditure within one or two parliaments; it could not. That any statute can bind a future Parliament absolutely; none can. That the historical figures reported here are settled: they are computed from a single published dataset and have not been independently corroborated. Or that this institution should be created: the paper offers a design conditional on the companion's economic finding, and does not demonstrate that the design would survive political time.

The thesis, and the problem the rest of the paper is built around. The decisive problem for a democratic state is preserving an investment institution through decades in which the opportunity cost of contributing is obvious but the eventual benefit remains largely invisible.

The tension it creates. The fund needs defenders now for a benefit that arrives later, but every pound spent on creating those defenders through present payments delays the benefit for which the fund exists.

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1. What the fund is for

State pension expenditure runs near 5% of GDP and the OBR projects it rising toward 9%. A fund capable of offsetting a meaningful share of that at a conservative drawdown would need to be measured in trillions.

There are two ways such a fund might contribute:

1. **Income.** Generate an annual distribution that offsets pension expenditure.
2. **Balance sheet.** Convert borrowing into a growing public asset that improves the state's net position.

These sound like two stages of the same project: accumulate an asset, then spend what it earns. **Whether they are compatible depends on how the fund is financed**, and that question has to be settled before any institutional design begins, because the answer determines what kind of institution is being designed.

2. The financial architecture

Four financial findings are load bearing for everything that follows. Each is stated here with the reasoning that makes it intelligible; the full test, the historical record, the sensitivity analysis and the falsification conditions are in the companion paper, and a reader who wants to dispute a finding should go there rather than here.

Finding 1. The income objective fails. The state pension bill grows at roughly 2.7% real, about 1.5% from GDP growth and 1.2% from the rising share of GDP the OBR projects. A distributing fund grows more slowly, so no plausible contribution path lets fund income offset a meaningful share of the bill. Reaching a tenth of it would need contributions of 0.6% to 1.2% of GDP sustained for fifty years.

Finding 2. The balance sheet objective succeeds. At £5bn a year and a 4.2% real return, from a global equity portfolio excluding domestic assets, the fund reaches roughly 13% of GDP by year 50 under the distribution rule section 7 adopts, and continues rising. Against a debt path the OBR calls unsustainable, an asset of that size is materially significant.

Succeeds here has a precise meaning, and the comparison it rests on is the one that matters. The fund is built with borrowed money, so the test is not whether it grows but whether it grows faster than the cost of the borrowing that created it. The relevant quantity is therefore the **net position**: the fund's value less the same contributions accumulated at the government's real borrowing cost. The objective succeeds when that difference is positive and rising.

A constant rate of return is not a constant contribution, and this is what makes the horizon intelligible. The 4.2% is a rate, not a quantity. Retained returns enlarge the capital on which the next year's return is earned, and continuing contributions enlarge it again, so the same percentage yields a larger sum every year: a mature fund produces a multiple of a young fund's annual real return without any improvement in its assumed rate. On the unchanged 4.2%, the fund's own annual return passes the annual contribution in year 18, stands at £5.9bn against £5bn by year 20 and £19.9bn by year 40, and from there more of each year's growth comes from accumulated capital than from that year's payment in. By year 50, on the unchanged 4.2%, the state is still contributing £5bn a year while the accumulated fund generates about £32.5bn from its own capital in that year, around six and a half times the annual contribution that built it. The stock figure carries the same point. With nothing ever distributed, £250bn of contributions produce a fund of about £812bn by year 50, so roughly £562bn of its value was

generated inside the fund rather than paid in by the state. That is the mechanism in its clearest form and not the path proposed here: under the distribution rule section 7 adopts, the year 50 fund is about £776bn, or 13.3% of GDP. The financing behaves the same way, which is why the test is the gap and not the fund. Both series compound, and the proposition is that the asset compounds faster.

On the historical record it is, and by a wide margin. Running the proposal against every complete rolling window since 1871 rather than a single assumed return path, **no forty year or fifty year window fails at the proposed distribution rate, and one thirty year window does.** That window begins in 1902 and contains the First World War, the collapse of 1920-21 and the onset of the Depression. Across 111 complete forty year windows of realised returns, on a flat share of realised real return distributed throughout, the fund could have distributed at least 13.2% of that return at the worst possible start, and around 49% at the median, while still ending ahead of its financing cost. The rule proposed in this paper distributes 5%.

Those figures are computed from a single published dataset and have not been independently corroborated. The construction, the sensitivity analysis and the conditions that would overturn them are set out in the companion paper.

Finding 3. Under debt financing the two objectives are mutually exclusive. Distribution stops the fund compounding while the debt beneath it keeps compounding, so a high distribution rule turns the net position negative between years 50 and 75, while a low distribution rule keeps it positive. The drawdown rule chooses between accumulation and income; it cannot deliver both.

Finding 4. Under surplus financing the trade off disappears, because no debt compounds underneath the fund. This is why the source of finance, not the investment strategy, is the binding constraint, and why section 8 is about funding models rather than a funding menu.

Two properties of the finance itself carry through to the governance design. The fund holds **no domestic assets**, following the Norwegian mandate, both because gilts held inside the public sector consolidate to nothing and because domestic holdings at scale would bid up domestic prices. That choice has since been tested against its strongest alternative rather than merely argued: the companion paper models a diversified world bond allocation directly, and finds that a sixty-forty portfolio fails three of 111 forty year windows where an equity-only portfolio fails none, because forty years of thinner compounding costs more than the hedge saves in one decade. **The confirmation carries a limiting condition that must travel with it.** The result holds on this paper's criterion, surviving over forty and fifty year horizons, and does not establish equities as generally superior. Should the institution later acquire liabilities falling due during downturns, the criterion changes and the allocation must be reconsidered rather than treated as settled. And the expected real spread over gilt financing is an equity risk premium bought with borrowed public money, which is what makes the leverage objection at section 8 the serious one. **Its size is contested, and nothing here rests on either figure.** A forward estimate from spot gilt yields gives roughly 0.4 to 1.2 points; the companion paper's realised record gives 3.68 to 7.40 points in every monetary regime since 1871, the narrowest occurring in the era of two world wars and the Depression. The discrepancy is unresolved, and the constitutional argument is deliberately built so that it does not depend on which figure holds.

Neither figure may be silently promoted into the financing assumption. The forward estimate and the realised record are both defensible and they disagree by a factor of three to six. Any later passage, or any reader, adopting one as settled is adopting a contested figure, and the direction of the temptation is predictable: since the realised spread is the more favourable of the two. Where a number is needed, both are stated and the choice is declared.

Everything numerical here is provisional pending independent corroboration of the underlying series. The historical back test specified in the companion paper has been run, and answers three of the four

questions a stochastic model was originally specified to address; the fourth, the probability of ever crossing into a positive net position, cannot be estimated from a record in which every window crosses. The net position result is sensitive to a poor early return sequence, because the debt compounds at a contractual rate while the asset base is still small, and the record bears that out: the one window that fails is the thirty year window noted above, over which the realised spread was 1.3 points against a record median above four, and it recovers at forty and fifty years.

That produces a design constraint rather than only a caveat. The evidence does not require that every constitutional provision operate on a forty year view; emergency powers, certification cycles and parliamentary review can properly act far sooner. What it constrains is any mechanism whose judgement implicitly asks whether the economic proposition has worked. **The founding architecture must not permit an adverse financial outcome inside the demonstrated minimum horizon, by itself, to constitute evidence that the institution has failed.** A thirty year economic reversion test would embed into the constitution precisely the sequence risk the constitution exists to survive. The distinction to hold is between institutional review, which may recur at any interval and asks whether governance functioned, contributions were made and the mandate was followed, and economic reversion, which may only rest on evidence measured over a horizon consistent with the proposition being tested.

3. Why this is not Norway

Norway is a drawdown comparator, not a funding comparator. Its rule permits spending around the estimated long run real return, currently put near 3%, and it exists because petroleum revenue was converted into financial assets while the state ran surpluses. The UK has neither the rent nor the surplus.

4. The failure modes

The comparative finding, stated as the section's thesis: **legal protection can make principal harder to spend, but it has not made future contributions unavoidable.**

- **New Zealand.** Established 2001 to partially pre-fund superannuation. Contributions suspended from 2009, resumed 2017. Statutory structure survived, assets stayed invested, nothing was raided. The fund's own accounts record that the suspension increased later required contributions and deferred withdrawals. This is the common failure mode and the one the design must defeat.
- **Ireland.** National Pensions Reserve Fund resources redirected into bank recapitalisation during the financial crisis. This is the dramatic failure mode.
- **Alaska.** Principal behind a constitutional amendment approved by voters, plus a visible citizen dividend. Survived fifty years of state fiscal cycles. But note the limit: the earnings reserve is spendable by legislative appropriation and now supports both dividends and public services. Alaska shows a protected principal and a visible benefit can survive. It does not show that a contribution path can be made unavoidable.

5. The constitutional position

The doctrine is stated rather than worked around. Under parliamentary sovereignty an Act cannot make itself unrepealable. A statutory contribution formula binds under current law; a later Parliament can amend, suspend or repeal it, and can repeal a referendum requirement by ordinary legislation before doing so.

Add the justiciability point, which is less obvious and more consequential. The closest precedent is the Sewel convention: Parliament placed it in section 28(8) of the Scotland Act 1998, and the Supreme Court in *Miller* nonetheless held it a political convention the courts would not enforce. A statutory duty on ministers to contribute is likely to meet the same fate. Trustees suing the Crown for a missed payment would in any case be a constitutional oddity.

The constitutional philosophy, stated once and plainly. The objective is not to prevent future governments making different choices. It is to prevent them making those choices cheaply. A future Parliament remains free to repeal the founding Act, dismantle the fund and spend the principal. What the architecture denies it is the ability to do any of that while pretending the decision has no consequence. Every mechanism in this paper forces consequences onto the public record. That is entirely consistent with parliamentary sovereignty, because it constrains the price of a decision rather than the power to take it.

Where that principle is weakest, conceded here rather than found later. Imposing political cost works when the cost is visible and falls on the person deciding. For a fifty year institution it frequently does neither. A Chancellor suspending contributions in year 8 bears a modest immediate cost and imposes a large one on 2070. Publication makes the distant cost visible, and the layer 2 requirement to report a missed contribution's value at the target date exists precisely for this, but visibility of a cost falling outside the political horizon is weak medicine. The citizen statement at layer 6 carries most of the weight against this failure mode, by making a distant cost salient now to people who will be alive to bear it. It is also the least tested of the six layers.

Conclusion of the section, which becomes the paper's governing question:

What institutional design would make suspending contributions legally conspicuous, fiscally transparent and politically expensive enough that governments do it only during genuine emergencies and subsequently restore the missed capital?

Absolute entrenchment is not answerable. That question is.

The asymmetry objection is met directly here rather than left to a reader. This paper asks Parliament to commit itself over decades on contributions while conceding that Parliament may change the fiscal framework whenever it wishes. The two commitments differ not in legal permanence, since neither has any, but in political salience. Changing the fiscal anchor is a constitutional act affecting debt measures, borrowing targets and Budget presentation, debated nationally because it alters the framework within which governments are judged. Skipping a contribution is an executive decision about one spending line. The external dependency cannot be eliminated, but it already sits behind a much higher threshold:

Parliament can always change its mind. The design objective is to ensure that changing the surrounding constitutional framework and abandoning the contribution path are both sufficiently visible, consequential and politically costly that they cease to be routine fiscal adjustments.

5A. Fiscal treatment and constraints

The architecture has to operate inside the fiscal framework by which governments are themselves judged. If contributions score as spending, or if converting cash into equities raises measured debt pound for pound, no constitutional design survives contact with a Budget. **The finding is broadly favourable, with three qualifications**, and an unfavourable one would have forced redesign rather than refinement.

The fund sits inside the public sector. ONS classified the National Wealth Fund within the central government subsector, citing public sector control over general corporate policy combined with constraints on both sides of its balance sheet. An endowment with a statutory purposes clause and government-appointed trustees would almost certainly be classified the same way.

Contributions are therefore not spending. As an intra public sector transfer, the contribution itself scores nowhere. What matters is the fund's acquisition of financial assets from the private sector, which is a financial transaction rather than expenditure and does not enter public sector net borrowing or the current budget. Contributions do not compete with hospitals inside the current budget. This is Treatment A.

The stock measure is decisive, and the 2024 rule change is what makes the proposal viable. Under PSND, only liquid assets are netted off, so converting cash into equities would have increased measured net debt pound for pound. Under PSNFL, targeted since October 2024, all financial assets and liabilities recognised in the National Accounts are included, including equity stakes. Borrowing to acquire portfolio assets is then broadly neutral. The proposal would have been arithmetically unattractive under the pre-2024 framework and is defensible under the current one.

And the mechanism is now observable in the aggregates rather than argued from principle. In July 2025 the ONS announced the comprehensive recording of the National Wealth Fund, the British Business Bank and the Development Bank of Wales inside the public sector boundary, revising the series back to 2014. The published impact is a reduction in public sector net financial liabilities of £0.7bn at March 2015, rising to £7.0bn at March 2025, attributed mainly to growth in the financial assets those institutions hold, including equity, loans and debt securities, against revisions to public sector net debt of up to £0.1bn. That is this section's argument demonstrated on real balance sheets: financial assets accumulating inside the boundary improve the targeted measure while barely touching the measure that excludes them.

Three qualifications:

1. **The cash requirement is real.** The net cash requirement rises and the DMO issues more gilts. No accounting treatment makes the financing free. The economic cost is the spread between gilt yields and expected portfolio return, which is the leverage argument under a precise name, and it is argued at section 8 rather than accounted around.
2. **The purposes clause must not defeat the market investor test.** Classification as acquisition of a financial asset depends on a genuine expectation of commercial return. A purposes clause that directs investment toward domestic policy objectives at below market returns risks reclassification as a capital transfer, which would score in borrowing. The governance design at layer 5 and the accounting treatment are therefore coupled, and the drafting must keep the investment mandate commercial.
3. **Consolidation cuts both ways.** If the fund is inside the public sector, its returns consolidate too and the drawdown is an intra public sector transfer. The fund never visibly pays for pensions in the accounts; it changes the composition of the balance sheet. Section 9's citizen statement can present an income attributed to pension cost relief, but that figure is a presentational construct rather than an accounting fact, and it is stated as such rather than claimed as an accounting fact.

5B. Specification dependencies

The constitutional objective is now defined, but several specifications determine whether an institution can be built to meet it, and they are unresolved. They divide into two classes, and the distinction matters: four determine whether the institution functions as intended, and one determines whether the surrounding fiscal system continues to recognise it in the intended way.

Internal dependencies

1. **The objective.** Offsetting the growth in pension expenditure, a specified share of its level, or general fiscal relief. Everything downstream follows from the choice.
2. **The contribution formula.** "£5bn a year" is a modelling assumption, not a rule. What revenue, sized how, charged on what, and indexed to what.
3. **The governance and suspension trigger.** The referendum requirement fails against rolling suspension unless the trigger is defined on cumulative unrepaired shortfall or consecutive years rather than on the label attached to the legislation.
4. **The drawdown rule and the legal character of the citizen interest.** A share of a single year's return produces unstable public income and pressure to distribute windfalls, so the measure must be smoothed. And if an individually legible share hardens into a property right, the fund becomes a defined contribution scheme and the proposal changes entirely. That is the same trap as an individual entitlement ledger, and the drafting must close it explicitly.

External constitutional dependency

1. **The fiscal metric and accounting framework.**

The proposal assumes continuation of the UK's present balance sheet aware fiscal framework. A future change to the fiscal aggregates against which governments are judged could materially alter the political viability of the contribution path without changing the legislation governing the fund itself.

This is not solvable by the fund's legislation, and no claim here suggests otherwise. It is declared rather than solved, and everything that follows proceeds on that stated basis. Hunt moved the target measure in 2022 and Reeves moved it again in 2024, which is the evidence for treating it as a live dependency rather than a background constant.

6. Governance: six linked layers

None sufficient alone. The claim is that together they close the cheap route to the money and make the remaining routes conspicuous.

The design brief, stated at the head of the section. A fifty year institution should not assume fifty years of wise ministers. It should assume ordinary governments acting under ordinary political incentives, and remain stable anyway.

Every layer below follows the same pattern: **remove discretion and increase transparency simultaneously.**

- certification removes ministerial discretion over the trigger while making it objectively visible;
- automatic reversion removes political bargaining while making the consequence predictable;
- shared liability funding removes the cliff edge decision while making the financing mix continuously observable;
- dual publication removes discretion over presentation while increasing what is published;
- the rolling governance metric removes interpretative volatility while the underlying accounts continue unchanged.

The cost, named here rather than left to a reader. Rules and transparency are usually alternatives: either you bind the decision, or you allow it and make it visible. This architecture does both,

and the price is rigidity. Every mechanism above removes the capacity to respond to a situation its designers did not imagine, and a fifty year institution will meet at least one. The answer is that the emergency exit exists at layer 1, through primary legislation stating the amount, the justification, the projected cost and a restoration schedule. The architecture is therefore not inflexible; it is expensive to bend, deliberately. That is stated plainly rather than presenting belt and braces as costless.

Layer 1. Statutory contribution formula. Calculated automatically from named receipts or fiscal measures, charged on the Consolidated Fund, reported but not recreated by the annual Budget. Suspension requires primary legislation stating the amount, the emergency justification, the projected long term cost, and a restoration schedule.

*The horizon this must survive, and why it is stated here rather than as a trigger. **Necessary because the companion paper measures two self-funding milestones a generation apart:** the fund's value exceeds the accumulated cost of the borrowing that built it at a median of ten years, while distributions become sufficient to finance the ongoing contribution at a median of thirty-two, and in the worst historical window the net position is not durably positive until year twenty-five.*

The gap between those two figures is the political problem this layer exists to address, and it should be stated in those terms rather than as a general appeal to patience. A Treasury asking when the fund overtakes its financing gets a decade. A public asking when contributions stop gets thirty-two years, which is longer than any parliament. **Both audiences will hear different answers to what sounds like the same question**, which is why both figures are given together above rather than left to emerge separately.

What this does not license. It does not create a horizon test, and no provision anywhere in this architecture should acquire one. The evidence establishes how long the commitment must survive; it does not license a rule that measures whether the fund has arrived. That distinction is the same one drawn at layer 3: the duration is a fact about what the institution must endure, not a fact about which any automatic consequence may turn.

Note what those four requirements have in common. **None prevents Parliament acting. All increase the price of acting.** They convert quiet non payment into an explicit parliamentary act, which is the only thing a statute can honestly attempt.

This is also the architecture's emergency exit, and the reason the rigidity conceded at the head of this section is a chosen cost rather than an oversight. It sits at layer 1 because only Parliament can legitimately override the settlement the later layers create. Ministerial waivers, Treasury directions or certifier escape clauses lower down would dissolve the hierarchy at exactly the point it claims to be strongest.

Layer 2. Contribution ledger. Any shortfall below the statutory path recorded publicly, controlling future contributions through a recovery formula once stated fiscal conditions return. Unlike a ledger recording deferred entitlement uprating, this need not be a recognised financial liability. The annual statement shows both the cash missed and its estimated value at the fund's target date. The second number is the politically meaningful one.

Layer 3. Independent reporting through existing machinery. A statutory duty on the OBR to publish cumulative unrepaired shortfall and its projected value at the target date in every Fiscal Risks and Sustainability report. No new watchdog, a figure the Treasury does not control, in a document already read.

The reporting basis is itself a durability instrument. A single year mark to market presentation manufactures the appearance of a fund oscillating around zero out of one that has crossed into a positive net position and held, and hands the liquidation argument to every downturn. The statutory basis is

therefore a rolling multi year net position, reported with three additions to what the OBR and the annual statement already publish:

- **the fund's own value and its drawdown from peak**, separately from any net position figure, because the first cannot be negative and the second can;
- **the fund's age**, stated beside the position, since the same position means different things at year 12 and year 32;
- **the position's location within the range of historically observed paths at that age**, which is the only form in which an early figure carries information at all.

The third is the operative one. It converts the question from *is the position positive*, which the evidence shows is close to uninformative early, into *is this fund behaving within the range of programmes that ultimately succeeded, at this age*. That question can be answered at year 12; the first cannot.

The position is evidence for judgement, not a statutory trigger. An earlier version of this paper treated the durability of a crossing as a measure of institutional credibility. The companion paper's historical testing shows that claim is too strong. Successful paths remain negative for decades and no discontinuity marks the age at which the position becomes decisive, so a minimum evidential age can be established and a reversion age cannot. **The rolling sustained position is reported to Parliament and supplied to the trustees as evidence bearing on the institution's condition. It produces no automatic constitutional consequence of its own.** The measurement is objective; the inference *measured position, therefore failed proposition* is not, which is why the indicator sits with the findings the trustees weigh rather than the class the OBR certifies.

The paper is not replacing accounting. It is creating a statutory governance measure, and the drafting must say so in terms, because the alternative reading is that the fund proposes to obscure losses. Annex A sets out the evidential basis for the floor, the terminology the statute must hold to, and the publication rule that keeps the two figures together.

Layer 4. Referendum reserved for constitutional alteration. Dissolution, expenditure of protected principal, or permanent abolition of compulsory contributions. Ordinary administration, investment policy and calibrated contribution adjustment remain with Parliament. **Critically**, the trigger must also fire on cumulative unrepaired shortfall or consecutive years of suspension, otherwise rolling temporary suspension achieves permanent abolition without ever meeting the threshold.

Layer 5. Independent trustees. With fiduciary duties and operational independence, and a statutory purposes clause against which their reports are written. **The purposes clause must make expected commercial return the governing investment principle.** Preferred sectors may be named only as a subordinate consideration. This is not a concession to markets; it is what the accounting treatment requires. Purposes such as supporting an industrial strategy or regional growth are politically attractive and, if they displace commercial return as the primary objective, weaken the argument that government is exchanging one financial asset for another. The architecture would then be undermining the fiscal treatment on which it depends.

The shape of the institution. Ownership and mandate with the Treasury under statute, trustees with fiduciary duties and a statutory purposes clause, and operational management contracted to a capable institution. The choice of that institution is subordinate to the classification test: the fiscal treatment set out above rests on central government classification, and assets held somewhere outside the targeted aggregate would dissolve the balance sheet purpose that section 2 found to be the only one that survives. **Whoever manages the fund must not also determine whether the statutory tests have been met**, which layer 3 gives to the OBR, and the benchmark must be set and the valuation audited outside the manager, because the drawdown and the reported position both depend on the fund's stated value.

What the trustees are for. The architecture separates eight judgements, from constitutional alteration down to portfolio valuation, and gives each to whoever is best placed to make that particular one. Those

judgements sit at three levels: operational execution, verification, and the constitutional question of whether the whole remains faithful to the statutory purpose. **The trustees occupy the third level only**, and perform none of the component judgements, because a body that also valued the assets or set the benchmark would become another operational institution and the level would stand vacant.

Every separation adds a seam, and the trustees are the only integrator. Their powers are therefore the most delicate specification in this layer: direct investment strategy and they become shadow managers, confine them to an annual report and they cannot maintain fidelity to purpose against anything. The middle ground is voice with procedural consequence and no operational decision. They may require explanations and remedial plans, publish to Parliament at a time of their choosing, **trigger** a formal parliamentary review rather than recommend one, and suspend **distribution only**, pending resolution of a finding. The suspensive power exists to preserve Parliament's ability to make an informed constitutional decision, not to substitute the trustees' judgement for Parliament's: it gives them time, not control. **Trustees are appointed by and removable only by Parliament, on fixed non renewable terms**, since an integrator appointed by the Treasury is a captured integrator and a renewable term gives its holder an interest in pleasing whoever renews it.

Annex B sets out the allocation of the eight judgements, the custodian question, the limits on the suspensive power and the case for separating valuation.

Layer 6. Citizen visibility. An annual UK Endowment Statement to every adult: total assets, assets per resident, the year's contribution, any shortfall, return, protected and spendable portions, and income attributed to pension cost relief. A legible share of a collectively owned national asset, expressly not a withdrawable account or an inheritable right.

Claim the right thing for it. The statement does not create accountability. It **lowers the cost of accountability**, which is a claim the paper can defend:

The state cannot guarantee that long term costs will become politically decisive. It can ensure they are made plainly visible to those who will bear them.

That promises the removal of informational barriers, not behavioural change, and it is a function the state already performs in many contexts.

Why this layer is different in kind. Layers 1 to 5 constrain government directly. Layer 6 enables citizens, and therefore works only indirectly, through democratic accountability. It rests on behavioural assumptions the others do not: that citizens read the statement, understand it, that it shifts political incentives, and that governments anticipate the shift. Those may hold. They are assumptions, and unlike the rest of the architecture they are not grounded in established constitutional practice. Whether the control is strong enough is uncertain; whether the state may legitimately equip it is not. The layer rests on the second claim, not the first.

Why it nonetheless carries weight. Political cost only prices behaviour when visibility, incidence and timing align. Layer 2 solves visibility mathematically and solves neither of the others: the minister deciding has left office by the target date, and many of those who bear the cost are too young to vote when the decision is taken. Layer 6 is the only mechanism addressing incidence and timing, which is why the least evidenced layer is also the one the intergenerational case depends on.

The referendum provision should state its own constitutional limit plainly:

The establishing Act would require any dissolution of the Fund, expenditure of its protected principal, or permanent alteration of its constitutional purposes to be approved by referendum. Under the doctrine of parliamentary sovereignty, a future Parliament retains the legal power to repeal or amend this requirement. The purpose of the provision is therefore not absolute legal entrenchment, but to raise the constitutional and political threshold for fundamental change.

What the six layers do not settle. Together they protect the flow into the fund: they make suspension conspicuous, its consequence automatic, and its reversal expensive. They say nothing about the flow out. Section 5 established that the fund is only a partial contributor to a named liability, with ordinary finance beneath it, so something must determine how much of the fund's return leaves it and when. That question cannot be answered by the machinery above, because a rule generous enough to build defenders defeats the accumulation the machinery exists to protect, and a rule austere enough to protect accumulation builds none at all. The next section takes it.

7. The drawdown rule

This is a constitutional choice, not an operational parameter. A low distribution rule says the state is building wealth. A high distribution rule says the state is smoothing fiscal income. Under debt financing the two cannot be combined, because every pound distributed stops compounding against a debt that does not stop compounding. Section 2 shows the rising schedule turning the net position negative between years 50 and 75, while the flat rule keeps it positive and growing. The companion paper's historical testing supports that choice with a measured margin: the maximum flat share of realised real return distributable throughout while the terminal net position remains positive is 13.2% at the worst forty year start and roughly 49% at the median, and a schedule distributing 60% exceeds the boundary at every start in the record.

That margin is evidence for the existing rule and not a licence to enlarge it. The test asked whether 5% survives the historical record; it did not ask what maximum extraction the constitution could tolerate, and the two questions have different answers for a reason the paper already gives. The distribution rate is a constitutional choice about whether the state is building wealth or smoothing income, and a comfortable empirical margin does not convert it into an optimisation problem. **The rule is unchanged. What has changed is the evidence beneath it.**

Time is one of the productive inputs into the institution. A pound distributed early is not merely a pound no longer held. It is the capital growth and reinvested return that pound would have generated across the remaining life of the fund. Distribution is not the same as a missed contribution. Skipping a contribution also skips the borrowing raised for it, so the debt leg falls alongside the asset; a distribution removes asset capital and leaves the financing already incurred untouched. Measured on that basis, and holding the debt leg identical, every £1 distributed under a rising schedule costs £3.15 of terminal net position. The distribution rule therefore protects future earning capacity as well as present capital.

The rule the paper adopts, following section 2's verdict:

- No ordinary drawdown before year 20.
- Years 21 to 30, up to 3% of realised real return.
- Thereafter up to 5% of realised real return, and no higher while any contribution is debt financed.
- Never a percentage of principal.
- Measured against smoothed expected real return rather than a single year's outturn, since an equity heavy mandate makes single year outturns unusable.
- **No increase in the spendable share while a contribution shortfall remains unrepaired.**
A government cannot stop paying in while increasing what it takes out.

The rising schedule to 50% or 60% of return is retained in the paper only as the rule that would apply under full surplus financing, where no debt compounds underneath the fund and the trade off dissolves. It should be presented as the alternative institution it creates, not as a later stage of the same one. **The choice between the two is financially load bearing rather than a governance preference.**

The same contributions and the same 4.2% give a year 50 fund of about £776bn under the rule adopted here and about £676bn under the rising schedule, and by year 75 the gap is 26.8% of GDP against 14.4%, because a schedule that distributes more as the fund grows takes its largest sums from the largest capital base.

The transition architecture

What follows is no longer a drawdown parameter. It is a transition architecture between two different purposes of the fund: entry deliberately difficult, continuation conditional, reversion externally certified, automatic and gradual, and the resulting reduction made administratively possible by tying distributions to a bounded liability capable of returning to ordinary finance.

Mixed funding makes historical attribution impossible: once surplus funded and debt funded contributions are in the same portfolio, no asset can be traced to either. Proportional attribution by cumulative contribution is economically clean and administratively opaque, which is the wrong trade for an institution whose central problem is durability.

The test should therefore be **prospective, not forensic**: is the state still borrowing to make net contributions to the fund? While it is, the balance sheet preserving rule applies. Once debt financed accumulation has ceased, Parliament may adopt the higher distribution institution.

That rule is a property of the institution rather than of individual contributions, which is what the paper's subtitle asserts. Three refinements are needed to make it work.

It converges on a current year fiscal test, and should admit that. Money is fungible, so a government running an overall deficit is borrowing to contribute whatever it calls the receipt. The prospective test is therefore in practice a test of the current fiscal position. The value of the prospective framing is not that it produces a different answer but that it supplies the right reason: the objection is to borrowing to build wealth while simultaneously spending that wealth down, not to being in deficit as such. Smooth it over a rolling period or determine it once a parliament, since an annual test would make distributions oscillate.

Close the lapse loophole. As drafted, ceasing to contribute satisfies the switching test and unlocks the higher distribution, which rewards abandoning the contribution path. The rule at layer 2 partly closes this, because ceasing creates an unrepaired shortfall against the statutory path. It closes fully only if the higher distribution additionally requires that the statutory path is being met from surplus, or has been completed or wound down by affirmative act rather than allowed to lapse.

Entry is constitutional; reversion is automatic. Adopting the higher distribution institution changes the fund's purpose and belongs with the referendum class decisions at layer 4. Reversion must not. If reversion also required affirmative approval, the architecture would become a one way ratchet: a government could continue elevated distributions through renewed debt financing simply by declining to start the reversal, and the referendum mechanism would have become a lock in device for the very spending pressure the governance layers exist to resist.

The authorisation to distribute at the higher rate should therefore be granted constitutionally but held conditional on the rolling financing test. If the condition ceases to be met, the fund reverts to the balance sheet rule automatically after a defined period. Distributions still fall, which remains politically difficult, but the burden moves: a government must legislate or seek renewed authority to prevent the reduction rather than choose actively to impose one. Entry changes the institution and requires democratic authorisation. Reversion restores the institution the founding statute created.

Two conditions make automatic reversion workable rather than merely stated.

It must step down, not cliff. A fall from 60% of return to 5% in a single year is a shock large enough that governments will legislate to prevent it, which would make the trigger formally automatic and practically elective. State the governing test rather than leaving a placeholder for the number. The glide must be:

- slow enough that the named liability can be returned to ordinary financing without immediate disruption; and
- short enough that renewed debt financing does not coexist with elevated distributions long enough for the balance sheet rule to become nominal.

Those conditions pull in opposite directions and the number sits between them. The natural anchor is the spending review cycle, since the ordinary finance share has to be absorbed through one, which suggests a glide of one to two review periods rather than a decade.

Someone other than the Treasury must certify the trigger. Automatic reversion is only automatic if the failure of the financing test is determined independently; otherwise "automatic" means only that the Treasury must declare a condition against its own interest, and the mechanism fails at the point it was meant to bind. The OBR duty at layer 3 should do double work: the body reporting the cumulative contribution shortfall is the natural body to determine whether the statutory path has been met, whether qualifying surplus finance continues, and whether reversion has been triggered, keeping the relevant facts together.

The drafting must preserve one distinction. **The OBR certifies the facts and the application of the statutory test. It does not decide whether reversion is desirable.** Parliament creates the rule, the certifier determines whether the published conditions have been met, and the consequence follows. Anything looser delegates a constitutional choice rather than an accounting determination.

The named liability is a condition, not a preference

The higher distribution should be **prohibited from funding general expenditure**, not merely encouraged toward a named liability.

Once fund income has disappeared into the aggregate spending settlement, reversion presents politically as a cut to government rather than as the restoration of another funding source, and the constitutional rule finds itself fighting the whole expenditure baseline. The most politically convenient use of the distribution would otherwise also be the use most likely to make automatic reversion fail.

A qualifying liability needs a defined annual call on fund income, an explicit ordinary finance fallback, and a prewritten reabsorption path for the glide.

One refinement to the fallback. It should be structural rather than contingent: the fund should be a partial contributor to the liability at all times, with the Exchequer as residual funder, rather than the sole funder with a fallback triggered on reversion. Reversion is then a change in the funding mix rather than a cut, which is administratively trivial and politically far cheaper. It also solves a mismatch the paper would otherwise carry, since fund income varies with markets while a demand led liability varies with caseload, and neither tracks the other.

Stated plainly: the earlier judgement, that a fixed 5% of return rule fails, was made against the income objective alone. Against the balance sheet objective it is the only rule that works.

8. Which funding models preserve the objective

Section 2 makes the source of finance the primary architectural constraint. The funding models can therefore be tested against the objective already chosen, and most of them ruled out.

Surplus financed. No debt compounds underneath the fund, both objectives are available, and the rising drawdown schedule becomes usable. Every pound is a net asset. This is the only model under which the fund can both distribute and accumulate. It is also the model a UK government is least likely to sustain.

Debt financed. The balance sheet objective remains available under a low distribution rule. The income objective does not. The fund's asset value and its financing adjusted net position must be reported separately: under debt financing it is the latter that shows whether accumulated investment performance has overtaken the accumulated financing cost.

Receipt financed. Asset sales, spectrum, carbon and windfall receipts sit between the two, and their classification depends on whether the receipt would otherwise have reduced borrowing. If it would, the contribution is economically debt financed whatever it is called.

On hypothecation, stated honestly: money is fungible and dedicating receipts is presentationally attractive but fiscally meaningless without a binding path. The formula at layer 1 does the work; the named sources make it legible.

On the leverage objection, directly: the fund is buying an equity risk premium with borrowed public money, which is the transaction the New Zealand assessment identified when it concluded the fund was effectively fully leveraged and taxpayers unlikely to be compensated for the risk. That objection does not depend on the size of the spread and is not answered by enlarging it.

The spread's size is contested, and the argument here deliberately does not turn on it. A forward estimate from spot gilt yields gives 0.4 to 1.2 points. The companion paper's realised record gives 3.68 to 7.40 points across every monetary regime since 1871, and finds no complete forty or fifty year window in which the proposition fails at the proposed distribution rate. An earlier version of this paper treated the thin forward estimate as itself a reason for constitutional protection: the economics were marginal, so an interruption could destroy a fragile proposition. **That justification is withdrawn.** It rested on one of two contested figures, and the evidence now available does not support describing the historical margin as thin.

What replaces it is the paper's own thesis, stated more precisely. The institution is not constitutionalised because the investment case needs nursing. It is constitutionalised because democratic incentives operate on a radically shorter horizon than the benefit being protected, and the companion paper now measures that gap: net position carries almost no information about eventual outcome for roughly the first fifteen years, and in the worst forty year window of realised returns, at the flat rule adopted here, does not become durably positive until year 25. A proposition can be economically robust across forty years and politically vulnerable throughout the decades in which that robustness remains invisible. **That is a stronger reason for the architecture than fragility was, and it is the reason the evidence supports.**

The spread must still be shown worth having over a century, or contributions restricted to genuine surplus. The leverage objection stands on its own terms.

A warning about the state in which that facility would be drawn, and about how it must not be described. The companion paper finds no general adverse relationship between equity returns and financing costs. The correlation is **positive**, at +0.291 across the record and +0.239 within deflationary years, and deflationary years deliver *better* than average real equity returns. What the record contains instead is a rare conjunction: four years in 150 combine a sharp real equity fall with a real financing cost above 5%, three of them belonging to one episode, and that episode produces the only thirty year failure in the record.

This must not be translated into deflation risk. Doing so would reproduce, in institutional language, a claim the companion paper tested and withdrew. The conjunction is not deflation-specific, since one of the four years occurred with inflation at +3.2%, and framing it as deflation points the design at an inflation hedge, which is the wrong instrument for a state that is not defined by inflation.

What the institution inherits is resilience to a conjunction, which is not an asset allocation problem and creates no new mechanism. It attaches to two requirements the architecture already has: the facility below, which exists to be drawn in exactly this state, and the duration commitment at layer 1, since the damage is done across the years of recovery rather than in the year of the fall.

And it carries a methodological warning this paper must observe in its own analysis. A stress test that varies financing conditions and asset returns independently cannot reach this state, however severe the individual shocks are made, because it explores only the space of independent departures. One decisive joint case establishes that; no claim about the general relationship is needed, and none is available. **Independent parameter stresses may not be presented as exhausting path risk.**

A liquidity facility must exist from foundation, and this follows from the evidence rather than from prudence. Necessary because the tested historical paths show the institution needing liquidity while the long-horizon proposition remains intact: in the worst forty year window of realised returns, at a five billion pound annual contribution and the flat distribution rule this paper adopts, the peak shortfall between the fund's value and the accumulated cost of the borrowing that built it reaches roughly £23bn, in a window that terminates strongly positive.

Two properties of that requirement matter more than its size.

It is foreseeable, so it is not an emergency. A shortfall that appears in the historical distribution is part of the mechanism rather than a failure of it, and a facility arranged after losses appear is a facility arranged at the moment of least political capacity. Establishing it at foundation converts a future crisis into a drawing.

Its existence follows from the evidence; its size and operating terms do not. The £23bn is one window's peak on one uncorroborated compilation, and the companion paper is explicit that it should not be treated as a design parameter. What the evidence establishes is that the facility must exist and must be capable of being drawn without a constitutional act, since requiring one would reintroduce at the moment of stress precisely the discretion the architecture removes elsewhere.

One quantity must not be mistaken for another here. The companion paper reports three figures in three dimensions: a peak shortfall of about £23bn, which is a stock and is the facility requirement; an exposure of about £65bn-years, which is an area under a curve and is not a sum of money; and a carrying cost under a billion pounds, which is what the second becomes once a financing rate is applied. Only the first sizes a facility.

9. The visibility problem

This section carries the central tension of the paper, and it is where the balance sheet reframing exacts its price. The old income framing had natural defenders: people who would one day receive money from the fund. The balance sheet framing has none, because improving the state's net worth is real but invisible, and an invisible benefit has no defenders in the year a Chancellor eyes the contributions. Section 5 conceded that political cost is weak medicine when the cost falls outside the political horizon. This section has to supply what carries the weight instead.

The problem stated precisely. A benefit defends a fund only if it is visible, if it accrues to identifiable people, and if those people are alive and enfranchised when the fund is threatened. The balance sheet benefit fails all three. It is a number in a statistical release, it accrues to the state in the abstract, and its largest beneficiaries are unborn. Every other layer constrains government. This layer has to manufacture defenders where the arithmetic has not provided any.

Why the obvious answer is wrong. The intuitive fix is a distribution large enough to be noticed, an Alaska style dividend. The arithmetic forbids it. At year 20 the fund is around £152bn on the central case, its real return around £6.4bn, and even distributing a fifth of that yields roughly £1.3bn nationally, which is a few tens of pounds a head, thinly spread and invisible against total managed expenditure, while removing the same sum from the compounding on which the whole proposition depends. Buying defenders with early distribution spends the fund's only advantage to purchase a benefit too small to defend it. The political yield is far weaker than it appears, and the compounding cost is exactly what the fund cannot afford in its early decades.

The sequence that follows from that. Visibility has to come from ownership rather than cash, until the fund is large enough that a real benefit costs little compounding.

1. Make ownership visible immediately, through the annual statement at layer 6, when the fund is small and the statement's only job is to establish the habit and the expectation.
2. Make skipped contributions visible immediately, through the ledger and the OBR duty, so the statement always reports what was contributed and what was withheld.
3. Introduce a small real benefit only once a statutory maturity threshold is passed, keyed to fund value rather than to a date, so the benefit cannot arrive before the fund can afford it.
4. Let the benefit grow only as a share of return the fund can distribute without ceasing to grow, which is the drawdown rule of section 7 doing double duty.

What the statement may and may not claim. Consolidation means the fund does not literally pay for pensions in the accounts: its returns are inside the public sector and net out. The statement must therefore read that the paper *attributes* a stated sum of the fund's investment income to pension cost relief, not that the fund *paid* that sum toward pensions. The first is a transparent reporting convention. The second is an accounting identity that does not hold, and a reader who knows the public finances would need only one sentence to discredit a paper that claimed it. The distinction is pedantic in ordinary language and decisive in front of the Treasury.

Two claims, kept separate because they sit in different epistemic categories. The section makes one conclusion and one hypothesis, and conflating them would let the hypothesis borrow the conclusion's certainty.

Established, as a conclusion of the arithmetic: if the architecture is to preserve long term compounding, it cannot buy political support through early distributions. This follows from section 2 and is not in doubt.

Hypothesised, and presently unevicenced: if ownership is made sufficiently visible, it may generate enough public attachment to raise the political cost of neglect before distributions begin. This does not follow from anything the paper has established.

Alaska calibrates the gap exactly. It demonstrates that a cash dividend can build a body of defenders. It says almost nothing about whether defenders can be assembled before any cash exists, which is the thing this layer needs. So the precedent is encouraging about the mechanism the arithmetic forbids and silent about the mechanism the arithmetic requires.

This is the least evidenced link in the architecture, and it is load bearing precisely because it is the only mechanism addressing the failure mode the others cannot reach. The discipline the corpus applies elsewhere applies here: the paper identifies the problem, explains why the tempting solution is incompatible with the arithmetic, proposes the least disruptive alternative, and then states plainly that the alternative's effectiveness is an evidence gap rather than an established fact. If future evidence shows ownership without early cash does generate attachment, the claim strengthens. If it does not, the architecture has correctly located the place a redesign would be needed rather than concealed it.

Why this gap will not close the way the quantitative one will. The modelling in the companion paper resolves to a number with a confidence interval, and a return sequence does not care whether anyone forecasts it. The behavioural claim resolves, at best, to "under these conditions, for these populations, for now." Worse, layer 6 alters the thing it measures. A statement that raises awareness alters political reporting, which alters parliamentary incentives, which raises visibility further; or the reverse runs and the statement becomes wallpaper nobody opens. Both are equilibria, and no reasoning *ex ante* picks between them, because the institution is an input to the behaviour it depends on. That reflexivity is the real reason this is harder than the modelling, and it means the honest register for layer 6 is not "unproven pending study" but "conditionally effective, self reinforcing or self defeating depending on an equilibrium the design cannot fully determine in advance." The claim made here is only that the mechanism is the least disruptive one compatible with the arithmetic, and that it is designed to be reinforceable if the favourable equilibrium begins to form: the maturity threshold and the later real benefit exist partly so that a defence of the fund, once begun, is given something to harden around.

10. What would falsify the proposal, and the two research agendas

The falsification tests divide by the kind of evidence they need, and the division is worth stating because the two agendas run in parallel and neither substitutes for the other.

Constitutional or design falsifiers, answerable by reasoning: - if no contribution formula can be specified that survives a single spending review; - if the citizen statement cannot be made legible without creating an enforceable individual claim; - if the OBR would not accept the reporting duty, or would report in a form that buries the number.

Quantitative falsifiers, answerable only by the modelling in the companion paper: - if sensitivity analysis shows the fund never reaching materiality under plausible return assumptions with achievable contributions; - if historical testing, or subsequent joint-structure-preserving testing, shows the fund failing to reach a positive net position by the end of the horizon on which the proposition is stated, under realistic return sequences.

This falsifier was rewritten. It previously read that the model would be falsified if the fund spent most of its first fifty years below zero, or crossed only briefly. Both halves rested on an interpretation of net position the companion paper's historical testing has since removed. A long early negative period is not by itself evidence of failure, since ultimately successful historical paths remain negative as late as year 33, and brief crossings that later reverse are normal enough in the early decades to carry no such implication, occurring in a quarter of paths at age 15. What can still falsify the model is terminal, not interim: a fund that does not reach a positive position by the end of its stated horizon has failed on its own criterion.

The behavioural falsifier, answerable by neither, and new to this paper: - if visibility of ownership without early cash does not generate durable defenders, layer 6 fails, and with it the only mechanism addressing costs that fall beyond the political horizon.

The corpus therefore leaves drafting with two external research agendas, not one. The Arithmetic of Patience carries the quantitative agenda. This paper carries a behavioural one, and it should be named as such rather than folded into the design questions it superficially resembles.

11. Implementation and sequencing

Legislate early in a parliament. Establish trustees and the reporting architecture before the first contribution, so the accountability machinery predates the money. First statement to citizens in year one, when the number is small, to establish the habit rather than to impress.

Annex: method, and relationship to the author's other work

A separate proposal of the author's on pension uprating is judged on its own merits, and any genuine saving it produces would be one candidate contribution stream among several. **Nothing here depends on it**, and stating that early prevents the objection that the proposals are being used to justify one another. This paper and its companion form a self-contained pair.

Carry across the methodology, but extend it. A specification test, asking which architectural choices are stated as settled when the mechanism achieving them has not been specified, identifies internal architectural dependencies. A second test is needed to identify assumptions about the institutional environment on which the architecture depends.

Internal specification test. Is there a sentence describing what a mechanism is meant to achieve, where the rule by which it achieves it has not been specified? Applied to a pension uprating proposal of the author's it found two such dependencies; applied here it found the citizen interest and the commercial mandate.

External dependency test. What assumptions about law, institutions, accounting, politics or behaviour does this mechanism rely upon? Who controls those assumptions? And can this paper alter them, or only declare them?

The third question matters most, because it prevents the design error of trying to solve inside the proposal something that lies outside its scope. The correct response to the fiscal metric dependency was not to invent legislation protecting PSNFL. It was to identify that the proposal is contingent on the continued use of a balance sheet aware fiscal framework, and to say so plainly.

The second test needs a threshold, or every environmental fact becomes a dependency and the category loses its force:

External dependency: an assumption about the institutional environment which, if altered, would materially change the operation of the proposal, and which the proposal itself cannot change except by declaring its reliance upon it.

The personal allowance remaining reserved qualifies, because devolving it would break a single UK-wide figure doing two jobs. That inflation exists does not, because the architecture is explicitly designed to operate across varying inflation outcomes. That future governments exist does not, because the governance layers already model it.

A third stage completes the method. **Boundary check:** separate unresolved legal, empirical or accounting questions from architectural ones. Compatibility with Convention rights is legal due diligence, not a design dependency. Whether an accrued balance scores as a present liability is an

accounting question, not a design dependency, even though the answer bears on the design. This stage prevents every uncertainty being promoted into a dependency, which would make the taxonomy useless by making it universal.

Stated as a sequence:

1. **Internal specification.** What rule makes the mechanism achieve the stated purpose?
2. **External dependency.** What condition outside the proposal must hold, who controls it, and can the proposal alter it or only declare it?
3. **Boundary check.** Is this architectural, or does it belong to legal advice, empirical modelling, accounting judgement or implementation?
4. **Resolution.** For each surviving issue, decide the disposition: specify it, declare it, investigate it further, or exclude it.

A note on where fixes are found. The four stages are diagnostic: they locate problems. They say nothing about where solutions live. That is a different kind of observation and it is not a fifth stage. Recorded as a heuristic, and deliberately not elevated beyond one:

When a mechanism resists satisfactory specification, suspect that the design problem lies in the relationship between institutions rather than inside the institution itself.

The pattern across both papers has been consistent, and not because relationships are inherently superior. The persistent failures were not caused by poorly specified objects. They were caused by responsibilities sitting in the wrong places:

- the certifier determines facts on which constitutional consequences depend, rather than making the constitutional judgement;
- the Exchequer permanently shares the funding of a liability rather than periodically replacing it;
- the drawdown rule governs the relationship between borrowing and distribution rather than setting a withdrawal percentage;
- the reporting basis governs the relationship between accounting and constitutional oversight rather than choosing an accounting convention.

There is not enough evidence to call this a method. There is enough to say it is where the successful redesigns have tended to come from.

A second pattern, closer to a principle than the first because the paper has arrived at it independently several times and reverted to it whenever it drifted:

No institution should both perform an action and certify the constitutional consequences of that action.

The certifier does not manage; the manager invests but does not judge statutory compliance; trustees guard the purpose but do not own the assets; Parliament changes the architecture but does not determine whether a statutory trigger has fired. Every time operational power and constitutional judgement were concentrated in one body, the design was later separated again. Worth noting as a recurring pattern rather than adopting as a formal rule, but it is one reason the architecture has grown more stable with each revision. Applied as a test rather than a description, it is also productive: it is what exposed the missing external valuation function at layer 5.

A note on how functions get assigned. The mature form of the question is not "what body should do this?" but "**what constitutional question must be answered, and which body is least**

conflicted in answering it?" Stable arrangements rarely depend on finding the perfect organisation. They depend on every necessary constitutional question having a clear answer, and on no single institution answering too many of them.

Least conflicted is necessary but not sufficient, because a body can be unconflicted and incapable. The OBR is unconflicted about valuation and has no capability to value a global portfolio. That test also assumes every constitutional function is indivisible, and valuation shows some are not. A prior question therefore comes first, stated as a design preference rather than a rule:

Where capability and independence are held by different institutions, first ask whether the constitutional function can be subdivided before accepting a compromise between them.

"What is the asset worth?" turns out to contain three questions: against what standard, what is the value under it, and was that correctly performed. Assigning all three to one body forces a choice between expertise and independence. Splitting them does not.

Two limits keep this from becoming dogma. **Decompose only where each component can be judged on its own terms.** Valuation splits cleanly because each of the three questions is independently assessable. Prudence of an investment strategy does not, because it is a property of the whole and splitting it into allocation and timing would produce two answerable questions and no answer to the one that matters. And **decomposition requires that some body retains accountability for the composite result without performing any component**, or the design achieves diffusion of responsibility rather than separation of judgement. Here that is the trustees, whose duty is fidelity to purpose and who perform none of the three valuation roles.

The design grammar, assembled

Stated together, because the individual observations have accumulated into something with a recognisable shape. Not a method, and not offered as one:

- separate judgements rather than institutions;
- decompose a judgement only where each component is independently answerable;
- retain accountability for the composite whole in a body that performs no component;
- resolve external dependencies before designing the institutions that depend on them;
- never let a body certify the constitutional consequences of its own action;
- make departures from the constitutional norm possible, but expensive;
- give every constitutional judgement an automatic consequence, without letting the judge choose the outcome.

The last governs the interaction between most of the others, and is better stated as a default presumption than as another item in the list:

Bodies determine facts or conditions. Statutes determine consequences. Parliament determines whether to alter the statute. Depart from that structure only where there is a positive reason.

It separates three things routinely collapsed together: fact finding, rule application, and political choice. The certifier performs the first, the statute the second, Parliament retains the third. Certification triggers reversion; a trustee finding triggers review; a missed contribution triggers a recorded shortfall; a failed

audit triggers correction. In each case the body answers only whether the specified condition exists, never what should now happen.

A corollary, not an exception. The presumption assumes the triggering judgement is factual, and constitutional judgements sit on a spectrum. Has the financing test been met, has the statutory contribution been made, was the valuation correctly performed: these are capable of objective determination, even where experts differ at the margins. Is the institution still faithful to its statutory purpose, does the architecture still embody Parliament's intent, has stewardship remained adequate: these cannot be reduced to measurement.

The degree of automaticity should be proportional to the objectivity of the triggering judgement.

Predominantly factual judgements may legitimately trigger automatic statutory consequences. Predominantly evaluative judgements should trigger a constitutional process rather than determine its outcome. That is why certification produces reversion while a trustee finding produces only a review.

Two consequences of the corollary.

The classification must be fixed in the founding statute, judgement by judgement. If where a trigger sits on the spectrum were determined at the time it fires, the classification would become the new locus of discretion, which is exactly what the architecture keeps removing. The spectrum is a drafting guide, not a live test.

It leaves the integrator structurally weak. The trustees make the most evaluative judgement and therefore carry the weakest consequence, while holding the broadest responsibility. That is the price of the corollary rather than an oversight, and it is the positive reason for the one compensating exception: a suspensive power over distribution only, which gives the integrator a lever proportionate to a review rather than to a reversion.

And one deliberate departure. Layer 1 has Parliament both asserting that an emergency exists and choosing the consequence, which breaches the presumption directly. The positive reason is that no other body can legitimately override a constitutional settlement. The departure is named here rather than left to pass, because an architecture that claims a presumption and quietly breaks it once is weaker than one that breaks it openly and says why.

A diagnostic for whether a constitutional test is load bearing. Good tests generate doctrine rather than requiring exceptions. The suspensive power's three limits are not independent safeguards: the duration rule prevents preservation becoming permanent control, the scope rule prevents it being exercised where no constitutional decision exists, and the shared funding rule prevents it coercing third parties. All three fall out of one proposition, that protective powers exist only to preserve Parliament's opportunity to make an informed decision. When several drafting rules emerge from a single test, the test is doing real work. When a test needs a carve out for each application, it is a label rather than a principle.

On the shape of the methodology note itself. It has stopped accumulating principles and started accumulating the conditions under which they cease to apply: the relational heuristic has evidential limits, the decomposition preference has conditions for decomposing and for preserving composite accountability, the sequencing preference applies only to external dependencies, and the presumption above distinguishes factual from evaluative triggers and names the circumstance in which it yields. Those conditions do not weaken the framework. They define its domain, and they let a reader decide not only when to use a principle but when not to.

Structural couplings. See the appendix. The architecture now contains pairs of provisions, introduced separately for unrelated reasons, that have become the constitutional justification for one another. Logical coherence means each mechanism follows from a principle. Structural coherence means mechanisms have become mutually necessary, which is a sign of maturity and a maintenance hazard in equal measure.

A note on the corpus, and the standard each paper asks to be judged by. Each should state its own standard, on the same principle that governs everything else here: declare rather than assume.

- a pension uprating reform of this kind succeeds if its constitutional reasoning is coherent and politically plausible.
- The Arithmetic of Patience succeeds only if the quantitative evidence supports its analytical claims, and fails honestly if it does not.
- This paper succeeds if its institutional design remains internally coherent under constitutional scrutiny.

Those are different standards, and a reader applying the wrong one will reach the wrong verdict. **The Arithmetic of Patience** asks what the long run arithmetic permits a sovereign balance sheet to become, and is judged on its evidence. **This paper** asks how institutions should be designed so that the arithmetic survives ordinary politics, and is judged on the coherence of its constitutional reasoning. A reader who has read one already knows how to read the other.

A sequencing rule follows from the external dependency test rather than from either pattern. **Resolve external dependencies before designing the institutions that depend on them.** The custodian question is the worked example: many papers would open by debating whether the central bank, an existing vehicle or a new corporation should manage the assets. The right order is to determine how the institution must be classified to achieve its constitutional purpose, and only then ask who is capable of managing it within that classification. Governance is downstream of classification, and designing it first wastes the work.

One further discipline, visible in the strongest revisions to both papers. It governs how objections are handled rather than how they are found:

1. identify the genuine objection, in its strongest form;
2. concede the part of it that is true;
3. show how the architecture accommodates the rest;
4. never claim a free lunch where there is a trade off.

The largest single change from the earliest drafts is that neither paper now argues its architecture is costless. Both argue that the costs are chosen, and smaller than the failures they are designed to prevent. That is a more defensible position for a constitutional paper, because it treats architecture as a series of explicit trade offs rather than an exercise in optimisation.

The fourth stage is what makes this a review process rather than a discovery process, and it acts on the paper's claims rather than only on its structure. Splitting a ledger question into its ownership and recognition halves moved one half from open to answerable, allowing a commitment on ownership and legal status while remaining explicitly contingent on fiscal recognition. Reducing uncertainty rather than reorganising it is the point.

Annexes

Annexes A and B carry the institutional specification supporting layers 3 and 5, Annex C the comparative evidence supporting section 4, and Annex D the clause shapes the architecture implies. The accumulation model, its assumptions and its sensitivities are not annexed here; they are the subject of the companion paper. They are not required to follow the argument. They are required to draft it, to audit it, or to dispute it in detail.

Annex A. The net position indicator: evidence, terminology and reporting

Supports layer 3. The main text states the reporting rule and the finding that the position is evaluative evidence rather than an automatic trigger. This annex sets out what that finding rests on, why it stops short of a trigger, and what the drafting must do to keep the evidence legible.

The evidence, and where it is tested. The companion paper profiles net position by fund age across every complete forty year window of the 1871 to 2020 record, at the proposed distribution rule, and reports four measures at each age. It rejects the hypothesis that year 25 marks the point at which position becomes decisive: no discontinuity occurs there on any measure, and informativeness develops gradually instead. Of paths showing a positive position at age 15, 26% later return to negative; at age 25, 11%; at age 30, 3%; by age 33, none. All 111 windows succeed terminally. The construction, the four measures and the limits of the sample are set out there, on the same single uncorroborated compilation as everything else drawn from that paper. What follows is what those results permit this institution to infer.

Why that establishes an evidential floor and not a trigger. Because the record contains no failed forty year path, the profile can say how often a successful programme still looks unsuccessful at a given age, and cannot say how often an unsuccessful appearance is real. **A minimum evidential age can therefore be established; a reversion age cannot.** Before roughly year 30 a negative position must not by itself constitute evidence that the economic proposition has failed. After that point it may become relevant evidence, and it remains insufficient to trigger an automatic consequence.

Where the provision sits in the paper's own grammar. A measurable fact may be certified; the interpretation of a measurable fact is a judgement; an automatic consequence is warranted only where the statutory fact is itself sufficient. Net position yields a number and therefore appears to belong on the automatic side, but its constitutional use requires the inference *measured position, therefore failed proposition*. The measurement is objective and the inference is not. The indicator accordingly moves to the evaluative side without being discarded, and joins the class of findings the trustees weigh rather than the class the OBR certifies.

A terminological requirement, and not editorial housekeeping. Necessary because the historical profile shows fund assets and net position telling materially different stories simultaneously: in the worst window the fund held £155bn against £115bn contributed while standing £23bn behind the accumulated cost of borrowing that sum. Once net position carries an evaluative constitutional role, keeping the two terms distinct is part of keeping the evidence legible. The statute and every reporting instrument must never describe the net position as *the fund*. An unleveraged portfolio cannot have a negative value, so a reader told the fund is in deficit has been told something impossible has happened. **The fund fell by £20bn and the fund's net position deteriorated by £20bn** describe different economic states, and only one of them can be true of a portfolio holding no leverage.

A governance measure beside the accounts, not instead of them. The two metrics answer different questions and are complementary rather than competing:

- accounts report the point in time position, because that is what accounts are for, and they continue unchanged;

- the governance measure asks whether the institution's balance sheet improvement has proved durable, because that is the constitutional purpose the fund was given.

There is precedent for a governance metric sitting beside the accounts without displacing them. The fiscal rules already work this way: PSNFL is a target derived from but distinct from the accounting aggregates, and the Whole of Government Accounts continue on their own basis alongside it.

The publication rule. One rule makes the distinction hold in practice rather than only in principle. **Both figures appear in the same document, from the same certifier, with the point in time position stated first.** If the rolling measure ever appears somewhere the point in time figure does not, the accusation of concealment lands, and it will land in the year the fund can least afford it.

Annex B. The allocation of authority

Supports layer 5. The main text states the purposes clause, the separation of manager from certifier, and the trustees' role as the only integrator. This annex sets out the custodian question, the full allocation of judgements, and the limits that make the suspensive power constitutionally defensible.

Where should the fund sit? The strongest precedent is Norwegian and it is close to what a UK reader would instinctively propose. Norges Bank carries out operational management of the Government Pension Fund Global through NBIM, a separate business area inside the central bank. But the separation is the point: the Ministry of Finance owns the fund and sets the mandate and benchmark, the Storting sets overall guidelines, and the Bank is agent rather than principal, operating under a separate Act and a separate management agreement from the one governing monetary policy.

Two UK specific risks would have to be resolved before adopting that home.

Classification. The fiscal treatment at the head of this paper rests on central government classification, following the ONS treatment of the National Wealth Fund. UK headline debt has at times been measured excluding the Bank of England. If fund assets sat inside the Bank and outside the targeted aggregate, the balance sheet purpose, which section 2 says is the only purpose that survives, would evaporate. This must be settled with the ONS before the custodian is chosen, not after. It is a specification question, not an administrative one.

Mandate conflict. The Bank sets Bank Rate and holds gilts. The fund's financing cost is the gilt yield and its asset values are rate sensitive. Norway carries the same structural tension and manages it by separation, helped by a foreign asset only mandate, which this paper has already adopted. But UK monetary independence is more contested than Norway's, and attaching a large equity portfolio to the institution that sets interest rates offers critics a new front and exposes the fund to the Bank's unpopularity in either direction.

A standalone statutory corporation on the New Zealand Guardians or CPPIB model is the alternative, and has the advantage of a purpose built mandate carrying no borrowed conflicts.

The allocation of functions. Eight judgements, and as little overlap as the design can manage.

Judgement	Holder	Question it answers
Constitutional change	Parliament	May the institution's purpose be altered?
Ownership and mandate	Treasury under statute	Whose asset, and on what terms?
Fidelity to purpose	Trustees with fiduciary duties	Is the institution still doing what it was created for?
Investment execution	Contracted manager	How is the mandate implemented?
Statutory conditions	OBR	Have the published tests been met?
Performance standard	Benchmark setter, outside the manager	Against what should the result be measured?
Portfolio value	Valuer	What is the asset worth under that standard?
Faithfulness of valuation	Auditor reporting to a parliamentary facing body	Was the valuation correctly performed?

The paper is not separating institutions for its own sake. It is separating **judgements**, and each belongs to whoever is best placed to make that particular one. That is more precise than a generic doctrine of separation of powers, and it is why the last three rows exist.

Three levels of judgement. The eight rows are not peers. They sit at three levels, and the distinction explains why the trustees matter more as the architecture grows more separated.

- **Operational:** execute the mandate, calculate valuation, apply the benchmark.
- **Verification:** audit the valuation, certify the statutory conditions, verify the reporting.
- **Constitutional:** taken together, are these activities still faithful to the institution's statutory purpose?

The trustees occupy the third level only. That is why they perform none of the component judgements: a body that also valued the assets or set the benchmark would become another operational institution and the level would be vacant.

The integrator problem. Every separation reduces conflict of interest and adds a seam. The architecture therefore runs two opposing tendencies: separate enough to avoid concentration, integrate enough to preserve coherent responsibility. The trustees are the only integrator, which makes their powers the most delicate specification in layer 5. Direct investment strategy and they become shadow managers. Confine them to an annual report and they cannot maintain fidelity to purpose against anything.

The middle ground is a body with voice and procedural consequence but no operational decision:

- require explanations from the manager, the valuer and the benchmark setter;
- require remedial plans and report on their execution;
- publish to Parliament at a time of their choosing rather than only on the annual cycle;
- **trigger** a formal parliamentary review, rather than recommend one.

That last verb matters. Recommendation leaves the outcome with the body being scrutinised, which is the standard failure of guardian institutions: responsibility without leverage. Making the review automatic on a trustee finding applies the pattern the paper already uses at layer 3, where certification produces a consequence without the certifier taking a decision. The trustees judge; the consequence follows; nobody has to agree with them for something to happen.

Two kinds of constitutional power. The architecture uses both, and they exist because the underlying judgements differ in kind rather than because one body outranks another.

- **Determinative.** A factual judgement causes a predefined statutory consequence. The OBR exercises this.
- **Protective.** An evaluative judgement cannot determine the outcome, but can prevent irreversible movement until the constitutional process has completed. The trustees exercise this.

The constitutional test the suspensive power must satisfy.

Its purpose is to preserve Parliament's ability to make an informed constitutional decision, not to substitute the trustees' judgement for Parliament's.

The suspension is not valuable because the trustees are right. It is valuable because distributions, once made, are difficult to reverse, and the pause preserves the status quo while Parliament considers a finding about fidelity to purpose. It gives the trustees time, not control. Elsewhere the architecture protects constitutional intent by removing discretion; here, where discretion cannot legitimately be removed, it protects intent by preventing irreversible change until the proper decision maker has acted.

Three consequences for the drafting follow from that test.

Duration is tied to the process, not to the calendar. The suspension lapses when Parliament disposes of the triggered review, or at a defined longstop, whichever comes first. Expiring before Parliament can consider the finding defeats the purpose; persisting after Parliament has decided converts time into control.

It is exercisable only in support of a triggered review, never standalone. Where no constitutional decision is pending, there is nothing for the pause to preserve, and a finding about operational quality is a matter for explanation and remedial plan rather than suspension.

It is proportionate only because of the shared funding rule. Because the fund is a partial contributor to a named liability with the Exchequer as residual funder, a pause changes the funding mix and harms no recipient. Had the fund been the sole funder, the same power would hold a benefit hostage to an evaluative judgement. The two design choices were made separately and neither works without the other, and the drafting should keep them together for that reason.

Appointment is part of the specification, not an administrative detail. An integrator appointed and removable by the Treasury is a captured integrator. Norway's Supervisory Council is appointed by Parliament. Trustees should be appointed by and removable only by Parliament, on fixed non renewable terms, since a renewable term gives the holder an interest in pleasing whoever renews it.

Why valuation is not one function. Valuation is an input to the constitutional machinery, not an operational task. The drawdown is a share of realised real return, and the sustained position indicator depends on the fund's stated value, so a manager that marks its own book influences both the operation of the statutory architecture and the evidence on which the trustees form a judgement. Re-founding the indicator as evidential does not weaken this argument; it broadens it. A number supplied for evaluative judgement requires independent valuation for the same reason a number carrying an automatic consequence does, because a body asked to weigh evidence cannot weigh a figure the manager has set. Norway separates it: external auditing is performed by the Central Bank Audit, reporting to the Supervisory Council appointed by Parliament, and NBIM's results are measured against externally defined benchmarks. The equivalent here is benchmark set outside the manager, valuation independently audited, and reporting to a body answerable to Parliament rather than to the manager or the Treasury. These are constitutional separations, not accounting refinements.

Annex C. Four sovereign-fund tests of the architecture

Supports section 4. The four comparators test different parts of the architecture rather than supplying four versions of the same warning. New Zealand shows why continuity of contribution matters even when principal is untouched. Ireland shows why protecting withdrawal is insufficient if political direction can alter the investment mandate. Alaska shows the significance of where constitutional protection ends, and its current reform supplies a live comparison with a move toward a constitutionally protected single fund and a capped draw. Norway supplies the mature steady state, while showing that successful accumulation eventually creates a different exposure as fund transfers become material to the fiscal position. Three funds therefore identify three distinct routes by which protection can be incomplete; the fourth shows both what successful maturation achieves and the dependency it creates. What the cases establish is that protection is multidimensional, not that the answer must have six layers: section 6 supplies that from the paper's wider reasoning. As section 3 notes, Norway is a drawdown comparator and not a funding comparator.

New Zealand: what interruption costs, when nothing is taken. Contributions to the New Zealand Superannuation Fund follow a statutory formula, and the Act permits the government to suspend them. They were suspended from 2009 until December 2017. The Guardians' own estimate, on a methodology agreed with the New Zealand Treasury and computed from published Treasury material, is around NZ\$12.5bn of missed contributions, around NZ\$12bn of returns forgone on those contributions, and around NZ\$3bn of forgone domestic tax on those returns. The fund was worth NZ\$37.5bn when contributions resumed. The protection of principal was never tested, because nothing was withdrawn. **The entire loss came from not paying in**, which is the failure layers 1 and 2 exist to make visible and expensive. Current fund value: primary-source figure pending.

Ireland: what a directed mandate costs, when the withdrawal rule holds. The National Pensions Reserve Fund was established to part-fund pension and public service costs from 2025, and could not be drawn on before then. From 2009 the Minister for Finance directed it into the two Irish banks: €20.7bn in total, €4.7bn into Bank of Ireland and €16.0bn into Allied Irish Banks. At 30 June 2015 that directed portfolio was valued at €13.6bn against a total fund of €21.2bn. The statutory bar on early withdrawal was not breached. **The fund was not raided; it was instructed what to buy**, and by 2015 the instruction accounted for nearly two thirds of everything it held. This is why layer 5 constrains the purposes clause and the mandate rather than only the exit.

Alaska: what protecting half a fund costs. Article IX, Section 15 of the Alaska Constitution protects the principal, which may be used only for income-producing investments. The Earnings Reserve Account that holds realised income is statutory, under AS 37.13.145(a), and is appropriable by simple majority. The fund stood at \$91.3bn at 30 June 2026; the draw is capped at 5% of a five-year average market value and limited to what the ERA holds, giving a FY27 draw of \$4.0bn against a five-year average of \$79.9bn. **Constitutional protection that covers part of a fund leaves a politically accessible account beside it**, and Alaska is now attempting to close it. A constitutional amendment before Alaskan voters would deposit the unencumbered balance of the earnings reserve into the fund as principal, applying to withdrawals from the fiscal year ending June 2028, and the corporation's published position is that a constitutional single-fund endowment would protect the fund against inflation, support a stable and sustainable annual draw, and extend constitutional protection. It does not characterise the two-account structure as a defect, and this paper does not attribute that to it: the institutional movement is the evidence. Alaska is therefore not only the limit case in section 4. It is a comparator moving toward the architecture this paper proposes.

Norway: the steady state, and its own exposure. The Government Pension Fund Global held NOK 22,683bn at 30 June 2026. The fiscal guideline ties spending over time to the expected real return, estimated at 3%, with withdrawals in a normal year kept well below that so the margin is available in a

downturn. About 27% of the 2026 central government budget is financed by transfers from the fund. That is the end state a balance sheet endowment reaches: perpetual capital supporting expenditure without routine liquidation. It is also a caution. **A fund whose transfers finance about 27% of the central government budget makes its own investment performance increasingly consequential for the fiscal position**, which is the condition section 9 describes in reverse, and it arrives only after the decades in which the institution is least visible and most vulnerable.

comparator	what was protected	what was not	quantified
New Zealand	the principal	the contribution	NZ\$12.5bn missed, about NZ\$12bn of returns forgone, on a fund then worth NZ\$37.5bn
Ireland	the withdrawal date	the mandate	€20.7bn directed, worth €13.6bn by 2015 against a €21.2bn fund
Alaska	the principal	the earnings account beside it	\$91.3bn fund, \$4.0bn FY27 draw, amendment pending to fold the reserve into principal
Norway	both, by rule rather than entrenchment	nothing yet	NOK 22,683bn, 3% guideline, about 27% of the 2026 budget

Canada and Sweden are omitted. Both are well governed and neither supplies a distinct failure or design lesson the four above do not already carry.

Annex D. Clause shapes for the contribution, suspension, ledger and referendum provisions

Supports section 6. These are shapes rather than drafting. They exist so that a reader can see what each layer would have to say in a Bill, and so that anyone who thinks a layer is unworkable can say which clause they object to. Parliamentary counsel would write these differently and better; what should survive their redrafting is the effect stated beneath each one.

1. The contribution. Layer 1.

There shall be charged on and paid out of the Consolidated Fund in each financial year a sum calculated in accordance with [the formula in Schedule 1], to be paid to the Fund. The sum shall be reported in the annual Budget documents but shall not require annual approval.

Effect: the contribution is a standing charge rather than an annual decision, so the default is payment and the act required is non-payment. The formula must attach to named receipts or fiscal measures so that the sum is calculable without discretion.

2. Suspension. Layer 1, and the architecture's emergency exit.

The sum payable under section [1] may be reduced or suspended for a financial year only by an Act of Parliament, which must state (a) the amount by which the sum is reduced, (b) the emergency justification relied upon, (c) the projected value of the reduced sum at the Fund's target date, as certified under section [4], and (d) a schedule for restoration.

Effect: none of the four requirements prevents Parliament acting; each raises the price of acting. The third is the one that does the work, because it converts a present saving into a published future cost. No ministerial waiver, Treasury direction or certifier escape clause may substitute for this section: only Parliament can legitimately override the settlement the later layers create.

3. The ledger. Layer 2.

Where the sum paid in a financial year is less than the sum calculated under section [1], the difference shall be recorded as an unrepaired shortfall. Where [the stated fiscal conditions] are met, the sum payable under section [1] shall be increased in accordance with [the recovery formula] until the recorded shortfall is extinguished.

Effect: recovery is automatic on a stated condition rather than a later political decision. The shortfall need not be a recognised financial liability; the annual statement reports both the cash missed and its estimated value at the target date, and the second figure is the politically meaningful one.

4. Reporting. Layer 3.

The Office for Budget Responsibility shall in each Fiscal Risks and Sustainability report publish (a) the cumulative unrepaired shortfall and its projected value at the Fund's target date, (b) the Fund's value and its drawdown from peak, stated separately from any net position, (c) the Fund's age, and (d) the Fund's rolling net position and its location within the range of historically observed positions at that age. The figures published under (d) are evidence bearing on the condition of the Fund and do not of themselves give rise to any consequence under this Act.

Effect: certification of a fact rather than a judgement, in a document already read, by a body the Treasury does not control. The final sentence is the whole of Annex A compressed into a saving provision, and it is what keeps a measured position from operating as a trigger.

5. Distribution. Section 7.

No sum may be paid out of the Fund before the twentieth year. In the twenty-first to thirtieth years the maximum distribution is 3% of realised real return; thereafter 5%. No distribution may be expressed as a percentage of principal. Realised real return shall be measured against smoothed expected real return rather than the outturn of a single year. The maximum distribution may not be increased while any shortfall recorded under section [3] remains unrepaired.

Effect: the rule is a share of return and never of capital, and the last sentence forecloses the combination that destroys the institution, which is a government that stops paying in while increasing what it takes out.

6. The referendum. Layer 4.

Any dissolution of the Fund, expenditure of its protected principal, or permanent alteration of its statutory purposes shall require approval by referendum. The requirement in this section applies equally where the cumulative unrepaired shortfall exceeds [threshold], or where the sum payable under section [1] has been reduced or suspended in [n] consecutive financial years.

Effect: the second sentence is the operative one. Without it, rolling temporary suspension achieves permanent abolition without ever meeting the threshold, and the referendum protects only the form of the institution while its substance is discontinued.

7. The saving provision, which must be in the Bill and not only in this paper.

Under the doctrine of parliamentary sovereignty, a future Parliament retains the legal power to repeal or amend this Act. The purpose of the provisions in this Part is not absolute legal entrenchment, but to raise the constitutional and political threshold for fundamental change.

Effect: the paper's central concession, stated in the instrument rather than about it. A statute that implied it had bound a future Parliament would be making a claim its own legal system rejects, and the first competent critic would say so.

Evidence still needed

- The New Zealand fund's current size, from its own annual report. The suspension cost, the comparator drawdown rules and the Irish and Alaskan figures have been obtained from primary sources.
- Confirmation from the OBR or the Treasury that an endowment of this kind would be classified as section 5A concludes. The classification question is answered there from ONS material and from the treatment now applied to comparable institutions; what remains is confirmation for this specific object rather than an open question about the framework.

Appendix: structural couplings

This appendix is written for future revisers rather than for readers. It is not needed to understand the architecture. It is needed to avoid dismantling it.

It is the second of two forms of institutional memory in this paper. The methodology note preserves the reasoning, so a later author can think in the architecture's terms. This appendix preserves the design, by recording where an apparently local improvement has non local constitutional consequences. Most papers defend an architecture against its critics. This one also has to defend it against well intentioned editors, since simplification is what later governments, officials and reviewers actually do.

Maintenance principle. Where two independently justified provisions have become the constitutional justification for one another, they are a coupled design. Amendment of either requires review of the other.

This is not a design principle. It exists to prevent a future editor concluding that a provision can be simplified away, without noticing that a second provision has silently lost its constitutional foundation.

Each entry below states the pair, why each side was introduced, and what breaks if either is removed alone.

1. Shared funding of the named liability, and the trustees' suspensive power. Shared funding was introduced so that automatic reversion would not create an intolerable cliff edge. The suspensive power was introduced so that an evaluative body could delay distribution without acquiring executive control. Remove shared funding and the suspension withholds a benefit from citizens on the basis of an evaluative judgement, which is a different constitutional act entirely. Remove the suspension and the integrator has responsibility without leverage.

2. The rolling reporting basis, and the evaluative sustained position indicator. The reporting basis was chosen so that a fund genuinely holding a positive position would not appear to oscillate. The sustained position indicator gives that rolling measure a defined interpretative purpose: it states what the rolling basis is for. Remove the rolling basis and the institution is judged annually on marks that manufacture the appearance the indicator exists to distinguish. Remove the indicator and the rolling basis has no stated purpose and reads as smoothing for its own sake, which is to say as concealment.

This pair was re-founded after the companion paper's historical testing. The earlier statement held that durability of crossing measures institutional credibility rather than investment performance, and that the two provisions were mutually dependent on that basis. The evidence does not support the first half. Durability becomes informative gradually, and the record cannot make it a classifier of failure. The dependency survives in a different and narrower form: the rolling basis prevents transient marks being presented as institutional condition, and the indicator supplies the interpretative purpose that keeps the rolling basis from reading as concealment. Because that interpretation is evaluative, it belongs with the trustees rather than with an automatic statutory consequence, and this pair therefore sits alongside pair 1 rather than in the certification class.

3. The classification test, and the custodian architecture. Classification was settled because the balance sheet purpose depends on where the assets sit in the fiscal aggregates. The custodian architecture was designed to separate ownership, mandate, management and certification. Choose a custodian before settling classification and the assets may sit outside the targeted measure, which removes the only purpose the analysis says survives. Settle classification and then disregard it in choosing a custodian, and the same follows.

4. The commercial mandate, and the fiscal treatment. The commercial mandate was introduced as a governance constraint on the purposes clause. The fiscal treatment as acquisition of a financial asset depends on a genuine expectation of commercial return. Soften the mandate toward policy objectives at below market returns and the transaction risks reclassification as a capital transfer, which scores in borrowing and defeats the architecture from the accounting side.

5. The decomposition preference, and composite accountability. Decomposition was preferred so that capability and independence could both be satisfied. Composite accountability was required so that separation did not become diffusion. Decompose without an integrator and every body discharges its mandate while nobody answers for the outcome. Keep the integrator but stop decomposing and it acquires the functions it is supposed to be judging, which vacates the constitutional level.

6. Automatic reversion, and independent certification. Reversion was made automatic so that a government would have to legislate to prevent a reduction rather than choose to impose one. Certification was assigned externally so that the trigger would not be determined by the body with an interest in the answer. Remove certification and reversion is automatic only in the sense that the Treasury must declare a condition against its own interest. Remove the automatic consequence and certification produces a published fact that obliges nobody.

Companion paper, data and licence

The companion paper. *The Arithmetic of Patience* tests the balance sheet proposition on which this architecture is conditional, reports the historical evidence, and states what would falsify each of its findings. Where this paper cites a figure drawn from that testing, the derivation, the declared analytical choices and the reproduction method are set out there. The two papers form a self-contained pair: neither requires any other work of the author's, and where a separate proposal is mentioned the point is stated so that it stands without it.

Data and attribution. The historical figures cited here are computed from the Jordà-Schularick-Taylor Macrohistory Database, release 6, consulted 13 August 2026, available at macrohistory.net, and made available by its authors at no cost under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International licence. In accordance with the attribution terms, the database is cited as follows, the first for the dataset generally and the second for its rate-of-return series:

Òscar Jordà, Moritz Schularick and Alan M. Taylor (2017), "Macrofinancial History and the New Business Cycle Facts", in *NBER Macroeconomics Annual 2016*, volume 31, ed. Martin Eichenbaum and Jonathan A. Parker, Chicago: University of Chicago Press.

Òscar Jordà, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick and Alan M. Taylor (2019), "The Rate of Return on Everything, 1870-2015", *Quarterly Journal of Economics* 134(3), 1225-1298.

The index behind the historical figures. They rest on an equally weighted index of the sixteen advanced economies for which that source publishes equity total returns, converted to sterling at contemporaneous rates from the same source and deflated by UK consumer prices. The two remaining countries in the database carry no equity series, so the sixteen are forced by coverage rather than selected. The construction and the choices declared with it are set out in the companion paper.

Fiscal aggregates are drawn from the Office for Budget Responsibility and used under the Open Government Licence v3.0. Contains public sector information licensed under the Open Government Licence v3.0.

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Status of the historical figures. They are computed from a single published source and have **not been independently corroborated**. A second compilation of the same quantities may differ, and a single source cannot reveal an error it contains throughout. They are therefore *measured and uncorroborated*: a weaker claim than established and a stronger one than estimated. The constitutional argument is written so that it does not depend on the precise values.

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